

Persian Medical Assistant Development Based on Large Language Models

Bachelor Project Presentation

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July 2025

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Chapter 1

Introduction

Persian Medical Assistant Based on Large Language
Models

Significant Progress in Large Language Models:

- Deep architectures and extensive text data enable natural language understanding
- Models like GPT, LLaMA, and T5 perform diverse tasks with just prompts
- Remarkable capabilities in general language tasks

Challenges in Specialized Applications:

- Limited access to organizational and up-to-date specialized data
- Risk of inaccurate or dangerous responses in sensitive domains
- Particularly critical in healthcare and legal applications

1.1 Project Objectives

Goal: Design and implement a medical language understanding assistant based on LLMs

Key Features:

- **Accurate Responses:** Specialized content-appropriate answers
- **Knowledge Base Access:** Internal organizational data integration
- **Safety Mechanisms:** Control and filtering for quality assurance
- **Domain Adaptability:** Healthcare, education, customer support

Three Main Components:

1. Fine-tuning with specialized medical data
2. RAG architecture for internal knowledge base
3. Control mechanisms for safety and accuracy

1.2 Dual-Track Project Structure

Two Parallel Development Paths:

Track 1: English Model

- English model with Persian support
- Fine-tuning on English medical data
- RAG implementation for bilingual environments
- Primary development path

Track 2: Persian Model

- Dedicated Persian model development
- Persian-specific fine-tuning
- Persian information retrieval
- Comparative evaluation

Enables comparison and analysis of both approaches

1.3 Report Structure

1. Chapter 2: Literature Review

- Research on LLMs for specialized applications
- Medical and organizational domain studies

2. Chapter 3: Model Fine-Tuning

- Dataset selection and preparation
- Base model tuning and evaluation

3. Chapter 4: RAG Architecture

- Internal knowledge base implementation
- Retrieval system design

4. Chapter 5: Security Mechanisms (Guardrails)

- Filtering methods and safety analysis
- Inaccurate response detection

5. Chapter 6: Conclusion

- Final analysis and future research suggestions

Project Significance

Key Contributions:

- First comprehensive Persian medical LLM assistant
- Novel dual-track development approach
- Integration of multiple state-of-the-art techniques
- Practical deployment-ready solution

Expected Impact:

- Improved medical information access for Persian speakers
- Framework for specialized LLM applications
- Advancement in multilingual medical AI
- Safe and reliable AI assistant for healthcare

Chapter 2

Literature Review

Research on Medical AI Assistants Based on LLMs

2.1 Dual Approaches

Initial Development:

- Most research based on English models with limited Persian support
- Challenges specific to Persian language and specialized data

Our Dual-Track Approach:

Main Track:

- English models with Persian support
- Leverage existing resources
- Cross-lingual transfer

Dedicated Track:

- Persian-specific development
- Native language optimization
- Comparative evaluation

Applied in both Fine-Tuning and RAG implementation

2.2 Medical Large Language Models

2.2.1 Foundation Architectures

- **Transformer & Attention:** Vaswani et al. - "Attention is All You Need"
- Foundation for GPT, PaLM, Gemma-3-12B-IT used in this project

2.2.2 Specialized Medical Models

- **Med-PaLM Series:** Singhal et al.
 - Clinical knowledge encoding
 - Expert-level performance (Med-PaLM 2)
- **MEDITRON:** Chen et al. - 70B model with medical pretraining
- **BioMistral:** Labrak et al. - Open-source medical LLM
- **Aloe Family:** Gururajan et al. - Fine-tuned health models
- **Smaller Models:** Meerkat, MedMobile for resource-constrained environments

2.3 Fine-Tuning Techniques

2.3.1 Supervised Fine-Tuning

- Brown et al.: Few-shot learning capabilities of LLMs
- Critical for medical applications with limited training data
- Foundation for supervised fine-tuning in this project

2.3.2 Resource Optimization Techniques

- **LoRA:** Hu et al. - Low-Rank Adaptation
 - Used for Gemma-3-12B-IT fine-tuning
 - Reduces trainable parameters
- **QLoRA:** Dettmers et al. - Quantized LLM fine-tuning
 - 4-bit quantization + LoRA combination
 - Enables large model training with limited resources
- **DLoRA:** Distributed parameter-efficient fine-tuning

2.4 Retrieval-Augmented Generation (RAG)

2.4.1 Fundamental Concepts

- Lewis et al.: First comprehensive RAG implementation
- Solution for improving accuracy and reducing hallucinations
- Foundation for RAG system in this project

2.4.2 Advanced Techniques

- **Graph RAG:** Edge et al. - Knowledge graphs for query-focused summarization
- **LightRAG:** Guo et al. - Simple and fast RAG solution

2.4.3 Novel Chunking Approach

- **Late Chunking:** Innovative technique implemented in this project
- Preserves full document context during embedding
- Significant improvement over traditional chunking methods

2.5 Persian Language Research

Current State:

- Limited research on Persian medical LLMs
- Most models focus on English with minimal Persian support
- Recent developments in Persian medical AI

Key Resources:

- **MF3QA Dataset:** First comprehensive Persian medical QA dataset
 - 20K training samples
 - Used for model training
- **Persian Medical Corpus:** For RAG system knowledge base
- Recent publication on Persian medical model training

Critical gap addressed by this project

2.6 Advanced Reasoning Techniques

2.6.1 Chain-of-Thought Prompting

- Wei et al.: Breaking complex problems into smaller steps
- Significantly improves reasoning capabilities

2.6.2 Extended Reasoning Methods

- **Self-Consistency:** Wang et al. - Improved chain-of-thought
- **Chain-of-Knowledge:** Li et al. - Linking with heterogeneous knowledge
- **Tree of Thoughts:** Yao et al. - Deliberate problem-solving approach

2.6.3 Medical Prompt Engineering

- Wang et al.: Comprehensive study on healthcare prompting
- Careful prompt design significantly improves medical task performance

2.7 Safety Mechanisms (Guardrails)

2.7.1 NeMo Guardrails

- NVIDIA's comprehensive framework for LLM control
- Content filtering and response accuracy assurance
- Preventing false information generation
- Conversation flow control
- Critical for medical applications

2.7.2 Specialized Security Patterns

- **Two-Layer Architecture:** Input & Output Guardrails
- **Input Control:** Filter inappropriate requests
- **Output Control:** Ensure response quality and safety
- **Medical Policies:**
 - Prohibit personal medical diagnosis
 - Control treatment recommendations
 - Emergency situation management

2.8-2.9 Training and Evaluation

2.8 Preference Tuning Techniques

- **RLHF:** Ouyang et al. - Human feedback training
- **DPO:** Direct Preference Optimization
- **KTO:** Kahneman-Tversky Optimization
- **ORPO:** Reference-free preference optimization

2.9 Datasets and Benchmarks

- **Medical Datasets:**
 - HEAD-QA: Complex health reasoning
 - MedMCQA: Large-scale multiple-choice
 - MedQA: Open-domain medical exams
- **Evaluation Metrics:**
 - MMLU: Multitask language understanding
 - PubMedQA: Biomedical research QA
 - BERTScore: Text generation evaluation

2.10 Multilingual Models

Bilingual Environment Challenges:

- Traditional word-frequency similarity limitations
- Solution: Multilingual embeddings with cosine similarity
- Cross-lingual information retrieval optimization

2.11 Tools and Technologies



Logos of key frameworks and tools used in the project.

2.12 Tools and Technologies

Core Infrastructure:

- **ElasticSearch:** Distributed search engine
- **Guardrails:** LLM constraint framework
- **LangChain:** LLM application framework
- **Ollama:** Local model deployment

Training & Evaluation:

- **Unsloth:** Optimized deep learning
- **TensorBoard:** Training visualization
- **Lightning.ai:** GPU research platform
- **Vast.ai:** Cloud training resources
- **lm-evaluation-harness:** Model evaluation

2.13 Summary and Key Innovations

Literature Review Findings:

- Medical AI assistants require intelligent technique combination
- Significant gap in Persian medical LLMs
- Need for safe and reliable medical AI systems

Project Innovations:

1. **Late Chunking Technique:** Enhanced RAG quality with full context preservation
2. **Two-Layer Guardrails:** Comprehensive safety in medical environment
3. **Optimal Technique Combination:** SFT + RAG + Guardrails integration
4. **Multilingual Solutions:** Persian-English environment optimization

Chapter 3

Supervised Fine-Tuning

Adapting Large Language Models for Medical
Applications

Part 1: English Training

- Dataset selection and preparation
- Initial model evaluation
- LoRA parameter optimization
- Dataset quality optimization
- Final training and evaluation

Part 2: Persian Training

- Motivation and necessity
- MF3QA dataset introduction
- Model selection criteria
- Training configuration
- Results and comparison

3.1 Introduction

Supervised Fine-Tuning:

- Key technique for optimizing LLMs for specialized applications
- Enables adaptation of pre-trained models to specific domains

Chapter Objectives:

- Adapt pre-trained models to medical domain
- Fine-tune four models: Aya-expense-8b, Gemma-2-9B, Gemma-3-12B-IT, gpt-oss-20b
- Train on both English and Persian medical data
- Optimize LoRA parameters for efficient training
- Evaluate and compare performance comprehensively

Part 1

English Training

3.2 English Dataset Selection

Dataset Quality Importance:

"Garbage in, garbage out" - High-quality training data forms the foundation of any successful model

Primary Dataset: UltraMedical

- Initial size: 410,000 medical instructions
- Comprehensive coverage of medical topics
- Filtered version: 200,000 high-quality samples

Table 1: Table 3.1: English Training Datasets

Name	Instructions	Filtered	In-dataset
MedQA	10.2K	9.3K	No
MedMCQA	183K	59K	Yes
MedQA-Eval	51.8K	51.8K	Yes
TextBookQA	91.7K	91.7K	No
PubmedQA	211K	88.7K	Yes
Chatdoctor	100K	31.1K	No
MedQuad	47K	6K	No
MedInstruct-52k	52K	23K	No
MedInstruct-120k	120K	25K	No
WikiInstruct	23K	23K	No
UltraMedical	410K	410K	–
Our dataset (filtered)	410K	200K	–

3.3 Initial Model Evaluation

Evaluation Methodology:

- Four models supporting Persian language evaluated
- Used lm-evaluation-harness package for standardized benchmarking
- Manual evaluation for Gemma-3-12b-it and gpt-oss-20b

Evaluation Datasets:

- HeadQA: 2,505 medical multiple-choice questions
- MedQA: 1,257 USMLE exam questions
- MedMCQA: 3,978 samples from 194K+ questions
- MMLU subsets: Anatomy, Clinical Knowledge, College Medicine, Medical Genetics, Professional Medicine

Initial Evaluation Results

Table 2: Table 3.3: Initial Model Evaluation Results (%)

Model	HeadQA	MedQA	MedMCQA	Anat.	C-K	C-M	MG	PM
aya-23-8B	37.21	31.62	31.52	39.26	41.77	45.40	40.40	34.83
aya-expanse-8b	44.67	34.69	34.46	34.44	51.81	43.15	42.42	44.57
Gemma-2-2b-it	37.72	29.12	31.07	31.85	41.77	36.30	37.37	32.21
Gemma-2-9b-it	58.00	46.66	42.08	51.11	60.64	62.33	60.61	60.30
Gemma-3-12b-it	70.97	65.06	55.23	65.80	75.10	75.39	78.82	78.64

Gemma-3-12b-it selected as primary model for English training

3.4 Initial Testing with Default Configuration

Initial Configuration:

- LoRA rank: 16
- LoRA alpha: 16
- Training data: UltraMedical (410K)
- Epochs: 1
- Learning rate: $2e-4$
- Batch size: 32

Results:

- Aya-expense-8b: <1% improvement
- Gemma-2-9b: <1% improvement
- Gemma-3-12b: 1.3% average improvement
- **Below expectations!**

Conclusion: Need to optimize LoRA parameters and data quality

3.5 LoRA Parameter Optimization

Optimization Methodology:

- Fast testing on Aya-Expanse-8B with 50K samples subset
- Testing rank/alpha combinations: 16, 32, 64, 128
- Evaluation on MedQA accuracy

Table 3: Table 3.7: LoRA Parameter Test Results

Rank-Alpha	(k,q,v,o)	(k,q,v,o,up,down,gate)	(all)
16-16	51.0%	53.7%	55.41%
32-32	50.6%	50.36%	52.7%
64-64	50.2%	50.0%	50.5%
128-128	48.21%	43.27%	42.40%

Key Finding: rank=32, alpha=32 selected for full training

3.6 Dataset Optimization

Data Filtering Process:

Filtering Criteria:

- Remove incorrect/ambiguous answers
- Remove duplicate questions
- Remove unreliable sources
- Remove off-topic content

Dataset Comparison:

- Original: 410K samples
- Quality score: 3.2/5
- Filtered: 200K samples
- Quality score: 4.1/5

Result: 51% reduction in size, 28% improvement in quality

3.7 Final English Training Configuration

Model Configuration:

- Model: Gemma-3-12B-IT
- Dataset: Filtered (200K)
- GPU: H100 SVM 80GB
- Batch size: 32
- Gradient accumulation: 2
- Epochs: 3
- Learning rate: $2e-4$

LoRA Configuration:

- Optimizer: adamw-8bit
- Weight decay: 0.01
- LR scheduler: linear
- Quantization: 4bit
- Target modules: all
- LoRA rank: 32
- LoRA alpha: 32
- LoRA dropout: 0

3.8 English Training Results

Table 4: Table 3.10: Performance Before and After Fine-tuning (%)

Model	HeadQA	MedQA	MedMCQA	Anat.	C-K	C-M	MG	PM
gemma-3-12b-it	70.97	65.06	55.23	65.19	75.10	75.35	78.79	78.66
gemma-3-12b-it-en-sft	77.33	74.28	61.46	70.38	80.72	81.65	82.82	81.65
Improvement	+6.36	+9.22	+6.23	+5.19	+5.62	+6.30	+4.03	+2.99

Achievement:

Average improvement: 7.6%

Final average accuracy: 77.3%

Comparison with Other English Medical Models

Table 5: Table 3.11: Comparison with Other English Medical Models

Model	Average Accuracy (%)
Yi-6B-Chat	54.8
Openchat_3.5	57.0
Mistral-7B-v0.1	56.3
Llama-3-8B-Instruct	63.9
Yi-9B	59.2
SOLAR-10.7B-Instruct	55.4
Medalpaca-7b	44.8
MMedLM2	55.8
BioMistral SLERP	53.3
Meditron-7B	30.8
Llama3-Aloe-8B-Alpha	66.0
PMC LLaMA 13B	54.6
gemma-3-12b-it-en-sft (ours)	77.3
Llama-3-70B-Instruct	76.7
Meditron 70B	52.1
Qwen1.5-72B-Chat	67.1
Yi-34B-Chat	66.8

Part 2

Persian Training

3.9 Persian Training Motivation

Necessity:

- Lack of specialized Persian medical models
- Limited Persian medical resources
- Need for culturally appropriate medical AI

Approach:

- Continue from English-trained models (transfer learning)
- Leverage English medical knowledge for Persian adaptation
- Address Persian-specific medical terminology and context

3.10 Persian Dataset: MF3QA

First comprehensive Persian medical QA dataset:

Dataset Composition:

- 20,000 training samples
- 2,000 development samples
- 2,000 test samples

Training Strategy:

- Merge train + dev = 22K samples
- Use test set for evaluation
- Focus on quality over quantity

Evaluation Datasets:

- Persian translations of MMLU medical subsets
- Same 5 categories as English evaluation

3.11 Persian Model Selection

Selection Criteria:

- Persian language support capability
- Acceptable performance in initial evaluations
- Good English training results (if previously trained)

Selected Models:

1. **gpt-oss-20b**: High capacity model with good Persian support
2. **Gemma-3-12b-it**: Best performer in English training

3.12 Persian Training Configuration

Key Differences from English Training:

- Much smaller dataset (22K vs 200K)
- Lower LoRA parameters to prevent overfitting
- rank=4, alpha=8 (vs rank=32, alpha=32 for English)

Gemma-3-12B Configuration:

- Dataset: MF3QA (22K)
- LoRA rank: 4
- LoRA alpha: 8
- Epochs: 3
- Batch size: 32

GPT-OSS-20B Configuration:

- Dataset: MF3QA (22K)
- LoRA rank: 4
- LoRA alpha: 8
- Epochs: 3
- Batch size: 64

3.13 Persian Training Results

Table 6: Table 3.15: Persian Model Performance (%)

Model	Anatomy	Clinical	College	Genetics	Prof.
gpt-oss-20b (baseline)	57	69	59	75	73
gpt-oss-20b-fa-sft	60	71	62	78	75
Improvement	+3	+2	+3	+3	+2
gemma-3-12b-it (baseline)	56	68	74	72	74
gemma-3-12b-it-fa-sft	58	70	76	74	76
Improvement	+2	+2	+2	+2	+2

Average improvement: 2.5% (acceptable given limited data)

Comparison with Other Persian Models

Table 7: Table 3.16: Comparison with Other Persian Models (%)

Model	Anat.	Gen.	Col.	Clin.	Prof.	Avg.
PersianMind	25.18	34.0	20.23	25.28	23.89	25.7
aya-expanse-8b	40.74	49.0	44.51	52.07	45.58	46.4
Qwen2.5	41.48	52.0	43.35	47.92	43.01	45.6
Gaokerena	48.18	53.0	43.49	55.47	47.22	49.47
gpt-oss-20b-fa-sft	60.0	78.0	62.0	71.0	75.0	69.2
gemma-3-12b-it-fa-sft	58.0	74.0	76.0	70.0	76.0	70.8

20% Higher accuracy over existing Persian models

3.14 Comprehensive English vs Persian Comparison

Table 8: Table 3.17: Training Comparison

Aspect	English Training	Persian Training
Training Data Size	200K samples	22K samples
LoRA rank	32	4
LoRA alpha	32	8
Batch size (Gemma)	32	32
Gradient accumulation	2	1
Average Improvement	7.6%	2.5%
Final Accuracy	77.3%	70.8%

Key Insight: Data volume significantly impacts achievable improvement, but careful parameter tuning can optimize results even with limited data

3.15 Chapter Summary

English Training Achievements:

- Evaluated three models: Aya-expanse-8b, Gemma-2-9B, Gemma-3-12B
- Optimized LoRA parameters (rank=32, alpha=32)
- Filtered dataset from 410K to 200K high-quality samples
- Final model **gemma-3-12b-it-en-sft**: 77.3% accuracy

Persian Training Achievements:

- Trained gpt-oss-20b and Gemma-3-12b-it models
- Used conservative LoRA (rank=4, alpha=8) to prevent overfitting
- Final model **gemma-3-12b-it-fa-sft**: 70.8% accuracy
- 25-30% improvement over existing Persian medical models

Key Contributions and Next Steps

Methodological Contributions:

- Demonstrated effective combination of:
 - 4-bit quantization
 - Optimized LoRA parameters
 - Data quality filtering
 - Multi-stage training
 - Language-specific parameter tuning

Final Models for RAG System (Chapter 4):

- **English:** gemma-3-12b-it-en-sft
- **Persian:** gemma-3-12b-it-fa-sft

Successfully adapted LLMs for medical domain in both languages

Chapter 4

Retrieval-Augmented Generation System Implementation

Bilingual Medical Knowledge Base Development

4.1 Introduction to RAG

Retrieval-Augmented Generation:

- Combines stored knowledge with text generation capabilities
- Effective solution for LLM limitations:
 - Access to up-to-date information
 - Reduction of hallucinations
 - Domain-specific knowledge integration

Chapter Structure:

- **Part 1:** English RAG system implementation
- **Part 2:** Bilingual system with Persian integration

Part 1

English RAG System

4.2 English Datasets

Table 9: English Datasets for RAG System

Dataset	Domain	Count	Chunking
Wikipedia	General	29.9M	Default
PubMed	Biomedical	23.9M	Default
Textbooks	Medical	125.8K	Default
StatePearls	Clinical	247K	Recursive + Late
MedCorp	Various	54.2M	
Final Selection	Medical	373.5K	-

Selected: StatePearls + Textbooks (quality over quantity)

4.3 Late Chunking Technique

Three-Phase Process:

Phase 1: Recursive Chunking

- Split by natural separators
- Size control with overlap

Phase 2: Simultaneous Vector Generation

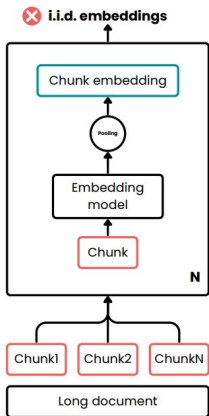
- Process all chunks together
- Preserve inter-chunk relationships

Phase 3: Aggregation and Pooling

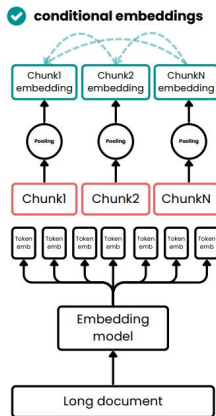
- Process entire document first
- Generate unified representation
- Split vector by boundaries

Key: Full context preserved in each chunk

Late Chunking Architecture



NAIVE CHUNKING



LATE CHUNKING

Traditional: Chunk → Embed → Store

Late Chunking: Full Document → Embed → Chunk → Store

4.4 English System Configuration

Chunking Parameters:

- StatePearls: 1024 chunk size
- Overlap: 100 tokens
- Method: Recursive + Late
- Textbooks: 300 chunk size

Storage Architecture:

- Elasticsearch backend
- Full-text search
- Vector search
- Hybrid search capability

4.5 English RAG Results

Table 10: English Model Performance with/without RAG

Model	HeadQA	MedQA	MedMCQA	Anat.	C-K	C-M	MG	PM
gemma-3-12b-it	70.97	65.06	55.23	65.19	75.10	75.35	78.79	78.66
gemma-3-12b-it-en-sft	77.33	74.28	61.46	70.38	80.72	80.83	82.82	81.65
+ RAG	78.83	75.42	62.20	71.56	82.10	83.12	82.50	82.46

Improvements:

+1.09% vs fine-tuned model

+6.73% vs base model

Highest gain in College Medicine (+2.29%)

Part 2

Bilingual RAG System

4.6 Persian Integration

Motivation:

- Lack of credible Persian medical resources
- Need to serve Persian-speaking users

Persian Medical Corpus:

- Comprehensive Persian medical content
- Collected from credible domestic sources
- Processed with Hazm library

4.7 Bilingual Dataset Overview

Table 11: Complete Bilingual RAG System Datasets

Dataset	Language	Original	Chunks	Method
Textbooks	English	125.8K	125.8K	Default
StatePearls	English	247K	247K	Recursive + Late
Persian Medical	Persian	102K	5,176,679	Recursive + Late
Total	Bilingual	-	5,550,179	-

Over 5.5 million chunks in bilingual system

4.8 Persian Processing

Preprocessing with Hazm:

- Text normalization
- Sentence boundary detection
- Appropriate tokenization

Chunking Configuration:

- Method: Recursive + Late Chunking
- Generated 5,176,679 chunks
- Max chunk: 500 characters
- Overlap: 100 characters

4.9 Bilingual Challenges

Cross-lingual Retrieval Challenge:

- Users ask in Persian, documents in English (and vice versa)
- Traditional methods (TF-IDF, BM25) fail in bilingual context

Solution:

Cosine similarity on multilingual embeddings

Benefits:

- Cross-lingual semantic understanding
- Unified knowledge base management
- Implicit concept translation

4.10 Persian RAG Results

Table 12: Persian Model Performance with/without RAG

Model	Anatomy	Clinical	College	Genetics	Prof.
gemma-3-12b-it (baseline)	56%	68%	74%	72%	74%
gemma-3-12b-it-fa-sft	58%	70%	76%	74%	76%
+ RAG	60%	71%	77%	75%	77%

Improvements:

+1.2% vs fine-tuned model

+3.2% vs base model

Notable achievement with 5M+ Persian chunks

4.11 System Performance Comparison

Table 13: English vs Persian RAG Performance

Metric	English	Persian
Chunks	373.5K	5.17M
Improvement vs Fine-tuned	+1.09%	+1.2%
Improvement vs Base	+6.73%	+3.2%
Best Domain Improvement	College Med (+2.29%)	All domains (+1-2%)

Key Achievement:

First comprehensive bilingual medical RAG system

4.12 Technical Limitations

Computational Constraints:

- Response latency from 5.5M chunks
- Storage requirements (15GB)
- Architecture complexity for bilingual management

Design Decisions:

- Excluded Wikipedia (29.9M) and PubMed (23.9M) due to size
- Focus on quality over quantity
- Optimized for medical domain specificity

4.13 Future Improvements

Technical:

- Graph RAG implementation
- Specialized medical embeddings
- Smart indexing optimization

Knowledge Base:

- More Persian medical sources
- Auto-translation of quality content
- Continuous updates
- Domain gap filling

Effective bilingual medical AI assistant achieved

Chapter 5

Security Mechanisms (Guardrails)

Ensuring Safety in Medical AI Systems

5.1 Introduction

Guardrails in Medical AI:

- Critical protective layer for medical assistants
- Prevents harmful, incorrect, or unethical content
- Ensures compliance with medical standards

Key Objectives:

- Input filtering
- Output monitoring
- Specialized medical policies enforcement

5.2 Medical Domain Challenges

Specific Challenges:

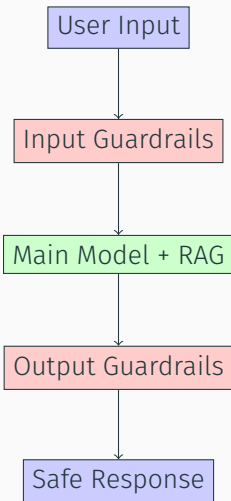
- **Unauthorized diagnosis:** System must not replace doctors
- **Drug prescriptions:** No specific treatment recommendations
- **PHI protection:** HIPAA compliance requirements
- **Emergency situations:** Proper escalation procedures
- **Drug interactions:** Preventing dangerous advice

Implementation Goals:

- User protection
- System credibility
- Legal compliance
- Risk management
- Trust building

5.3 Guardrails Architecture

Multi-layer Architecture:



Key Components:

5.4 Input Security Policies

Nine Key Input Controls:

1. No personal diagnosis requests
2. Protection of safety rules
3. Emergency situation management
4. Sensitive content filtering
5. Abusive language prevention
6. PHI protection
7. Identity impersonation prevention
8. System security
9. Malicious code blocking

Blocks inappropriate requests before processing

5.5 Output Security Policies

Eight Core Output Criteria:

- **Harm prevention:** No dangerous medical advice
- **Professional disclaimer:** Not a doctor replacement
- **Treatment limitations:** No specific recommendations
- **Information accuracy:** No medical misinformation
- **Self-diagnosis prevention:** No encouragement of self-treatment
- **Evidence-based:** Avoid unverified claims
- **Appropriate tone:** Clear, empathetic, cautious
- **Proper referrals:** Direct to professionals when needed

Ensures safe and appropriate responses

5.6 Technical Challenges

Implementation Challenges:

- Balance between safety and usability
- 176% increase in response time (additional 730ms)
- Comprehensive scenario coverage
- Bilingual support (Persian/English)

Current Limitations:

- Context detection challenges
- Increased computational load
- Need for continuous policy updates
- Cultural adaptation requirements

5.7 Optimization Strategies

Performance Optimization:

- Lighter specialized models
- Parallel processing
- Result caching
- Rule-based filtering for obvious cases

Accuracy Enhancement:

- Continuous learning from feedback
- Threshold optimization
- Multi-technique combination
- Adaptive sensitivity

Goal: Reduce latency while maintaining safety

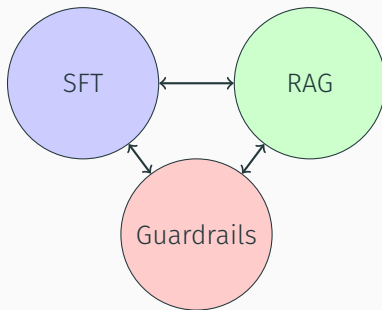
Chapter 6

Conclusion and Future Work

Project Summary and Recommendations

6.1 Project Summary

Comprehensive Medical AI Assistant Development:



Integrated Bilingual System

Three core technologies integrated for reliable medical assistance in English and Persian

6.2 Key Achievements - English

Fine-tuning Results:

- Model: Gemma-3-12B-IT with 200K medical samples
- Best improvement: MedQA with +9.22% accuracy
- Final model: gemma-3-12b-it-en-sft

RAG System:

- 373.5K specialized documents (StatePearls + Textbooks)
- Late Chunking technique implementation
- Additional improvement: up to 2.29% in some domains

6.3 Key Achievements - Persian

Model Development:

- Two models trained: gpt-oss-20b and Gemma-3-12B-IT
- Dataset: MF3QA (22K samples)
- Best model: gemma-3-12b-it-fa-sft (70.8% accuracy)

Bilingual RAG:

- Persian Medical Corpus: 5,176,679 chunks
- Cross-lingual search capability
- Comprehensive bilingual platform

6.4 Final Performance Summary

Table 14: Overall System Performance

System	Base	Final	Improvement	Avg. Accuracy
English	70.54%	77.27%	+6.73%	77.27%
Persian	68.8%	72.0%	+3.2%	72.0%

Strengths:

- Comprehensive improvement across all metrics
- High security with content filtering
- Full bilingual support
- Cross-lingual search capability

Weaknesses:

- 176% increase in response time
- Limited English RAG data
- Imbalanced bilingual resources

6.5 Practical Applications

Clinical:

- Medical centers
- Online consultation
- Decision support

Educational:

- Medical training
- Student platforms
- Public health literacy

Research:

- Literature access
- Cross-lingual collaboration
- Knowledge synthesis

6.6 Future Work - Priority 1

Reinforcement Learning Implementation:

- Improve response quality through user feedback
- Adapt to user preferences
- Reduce inappropriate responses
- Ensure balanced bilingual quality

Implementation Plan:

1. Collect bilingual expert feedback
2. Design reward function (accuracy, safety, usefulness)
3. Apply PPO or DPO techniques
4. Separate evaluation for each language
5. Continuous improvement system

6.7 Future Work - Priority 2

Guardrails Optimization:

Target: Reduce response time to <500ms while maintaining >90% filtering accuracy

- **Smaller models:** DistilBERT (English), ParsBERT (Persian)
- **Parallel processing:** Async pipeline optimization
- **Smart caching:** Semantic hashing for similar requests
- **Complexity reduction:** Rule-based for simple cases

RAG Enhancement:

- Fine-tune embedder for medical domain
- Implement Graph RAG
- Expand Persian knowledge base
- Balance bilingual resources

6.8 Final Conclusion

Project Achievements:

- ✓ Successful bilingual medical assistant development
- ✓ 5.5M+ chunk RAG system implementation
- ✓ Comprehensive security mechanisms
- ✓ Proof of concept for Persian medical AI

Key Contributions:

- First comprehensive Persian medical AI system
- Novel Late Chunking implementation
- Effective integration of SFT + RAG + Guardrails
- Template for other specialized domains

Ready for real-world clinical and educational deployment

Questions?